

Detailed Project Proposal: Real-Time Multi-Target Tracking Using YDLIDAR G2 and Unscented Kalman Filter with CTRV Motion Model

Course Reference: EE4084 Project

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Abstract

This extensive project proposal outlines the design, theoretical foundations, implementation strategies, and evaluation metrics for a robust, real-time multi-target tracking system operating in dynamic and cluttered indoor environments. The proposed system directly references and expands upon the framework established in the file named **EE4084_Project_Paper.pdf**. By capturing raw spatial measurements via a 2D YDLIDAR G2 sensor, the framework implements a Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to segment and isolate potential moving targets. State estimation under non-linear target dynamics is handled using an Unscented Kalman Filter (UKF) operating on a Constant Turn Rate and Velocity (CTRV) motion model. Measurement-to-track association is sustained via a nearest-neighbor gating strategy with explicit track management logic to prevent false track initializations and manage temporary occlusions. Finally, the statistical accuracy and filter consistency are strictly monitored utilizing Normalized Innovation Squared (NIS) metrics validated against Chi-square (χ^2) confidence bounds.

1 Introduction and Motivation

In the field of autonomous systems, robotics, and intelligent surveillance, the ability to perceive, detect, and track multiple moving targets simultaneously in real time represents a fundamental requirement for decision-making and collision avoidance. Sensors like cameras provide rich visual context but suffer from illumination sensitivity and lack direct, high-accuracy depth measurement capabilities. Conversely, Light Detection and Ranging (LiDAR) sensors provide exceptional spatial accuracy and precise distance mapping regardless of light variations, making them highly desirable for dynamic indoor environments.

The primary challenge in multi-target tracking (MTT) resides in managing measurement uncertainties, environmental clutter, sensor noise, and complex target maneuvers[cite: 1]. Traditional linear estimation frameworks, such as the standard Kalman Filter (KF), are structurally restricted to linear process and measurement equations. To mitigate non-linear constraints, the Extended Kalman Filter (EKF) utilizes first-order Taylor series approximations to linearize the system around the current state estimate[cite: 1]. However, EKF models introduce significant linearization errors when targets undergo high-rate non-linear maneuvers, often causing filter divergence[cite: 1]. While Particle Filters (PF) offer robust alternatives for arbitrary non-linear densities, their computational overhead presents severe limitations for low-latency, real-time systems running on standard computational platforms[cite: 1].

To bridge this gap, as documented in **EE4084 Project Paper.pdf**, this project utilizes an Unscented Kalman Filter (UKF)[cite: 1]. The UKF addresses non-linear propagation by choosing a deterministic set of sample points (known as sigma points) that perfectly capture the mean and covariance of the state distribution[cite: 1]. When these points are propagated through the true non-linear system, the posterior mean and covariance are accurately captured up to the third order of Taylor series expansions without requiring explicit, cumbersome Jacobian matrix computations[cite: 1].

2 Literature Review and Related Work

State estimation and tracking frameworks using LiDAR point clouds have seen extensive development. A notable benchmark by Sualeh and Kim[cite: 1] integrated an Interactive Multiple Model Unscented Kalman Filter with a Joint Probabilistic Data Association Filter (IMM-UKF-JPDAF) for dynamic multiple object detection using multi-LiDAR setups[cite: 1]. Their architecture showed that leveraging advanced data association combined with non-linear filters drastically decreases false-track propagation in highly cluttered spaces[cite: 1]. Similarly, studies focusing on intelligent transportation systems have successfully utilized roadside and vehicular LiDAR networks to monitor urban traffic trajectories via adaptive Kalman schemes[cite: 1].

While many high-end systems rely on complex 3D grid-based voxel segmentations and multi-layered sensory arrays[cite: 1], this project focuses on optimizing real-time tracking from a single, budget-friendly 2D LiDAR (YDLIDAR G2)[cite: 1]. By matching a lightweight spatial clustering pipeline with a highly stable CTRV-based UKF model, this project ensures high localization accuracy while maintaining a low computational footprint suited for standard embedded applications[cite: 1].

3 Proposed System Architecture

The comprehensive software and hardware processing pipeline of the proposed tracking system is structured into four sequentially execution modules:

1. **LiDAR Data Acquisition:** Interfacing with the YDLIDAR G2 physical layer to stream raw range and angle measurements[cite: 1].
2. **Preprocessing and Spatial Clustering:** Converting raw data into Cartesian coordinates, eliminating static ambient clutter, and applying DBSCAN to extract unique target centroids[cite: 1].
3. **Recursive State Estimation (UKF):** Propagating and updating target states using a deterministic sigma-point framework under non-linear CTRV kinematics[cite: 1].
4. **Track Management and Statistical Validation:** Building measurement-to-track gating boundaries, updating lifetime flags for tracks, and verifying estimation stability via NIS analysis[cite: 1].

3.1 LiDAR Preprocessing and Coordinate Conversion

The YDLIDAR G2 sensor captures local surroundings in a polar coordinate format, outputting pairs of measured distance (r) and scan angle (θ)[cite: 1]. To execute Euclidean clustering, these vectors are projected onto a 2D Cartesian plane using the fundamental trigonometric mappings[cite: 1]:

$$p_x = r \cos(\theta) \quad (1)$$

$$p_y = r \sin(\theta) \quad (2)$$

Following projection, a strict range filter is enforced[cite: 1]. Points residing outside the predefined spatial boundaries or representing zero-reading reflections are removed instantly to reduce downstream processing loads[cite: 1].

3.2 Target Clustering via DBSCAN

Rather than using heavy 3D voxelization or rigid grid maps, clustering is executed via the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm[cite: 1]. DBSCAN groups points based on spatial proximity without demanding a prior assumption of the exact number of clusters[cite: 1]. The algorithm relies on two parameters[cite: 1]:

- **Neighborhood Radius (ϵ):** Set empirically to 0.25 meters[cite: 1].
- **Minimum Points ($MinPts$):** Maintained at a minimum of 5 points[cite: 1].

A coordinate point P is designated as a *core point* if at least $MinPts$ reside within its ϵ -ball[cite: 1]. Clusters are built by iteratively checking density-reachable properties[cite: 1]. Points failing to fulfill density criteria are rejected as ambient noise or static background structures[cite: 1]. For every isolated dense cluster, its geometric centroid is computed and transferred to the state estimator as an official positional observation (z_k)[cite: 1]:

$$z_k = \begin{bmatrix} p_{x,meas} \\ p_{y,meas} \end{bmatrix} = \begin{bmatrix} \frac{1}{M} \sum_{j=1}^M p_{x,j} \\ \frac{1}{M} \sum_{j=1}^M p_{y,j} \end{bmatrix} \quad (3)$$

4 Mathematical System Modeling

The tracking framework is modeled as a non-linear discrete-time state-space system governed by the following state transition and measurement equations[cite: 1]:

$$x_k = f(x_{k-1}) + w_k \quad (4)$$

$$z_k = h(x_k) + v_k \quad (5)$$

where $w_k \sim \mathcal{N}(0, Q)$ denotes the Gaussian process noise accounting for motion modeling uncertainties, and $v_k \sim \mathcal{N}(0, R)$ represents the additive sensor measurement noise[cite: 1].

4.1 The Constant Turn Rate and Velocity (CTRV) Model

To model dynamic human trajectories inside an indoor lab environment, the target state vector at any given discrete epoch k is selected as a five-dimensional representation[cite: 1]:

$$x = [p_x \ p_y \ v \ \psi \ \omega]^T \quad (6)$$

where p_x and p_y indicate spatial positions, v denotes scalar linear velocity, ψ describes the instantaneous heading direction, and ω represents the angular turn rate ($\dot{\psi}$)[cite: 1].

Assuming the turn rate and velocity components remain constant during a sampling interval Δt , the continuous-time derivative of the state vector is given by:

$$\dot{x} = \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{v} \\ \dot{\psi} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \cos(\psi) \\ v \sin(\psi) \\ 0 \\ \omega \\ 0 \end{bmatrix} \quad (7)$$

Integrating this derivative over Δt for a non-zero turn rate ($\omega \neq 0$) yields the explicit non-linear state propagation function $f(x)$ [cite: 1]:

$$p_x(k+1) = p_x(k) + \frac{v}{\omega} (\sin(\psi + \omega\Delta t) - \sin(\psi)) \quad (8)$$

$$p_y(k+1) = p_y(k) + \frac{v}{\omega} (-\cos(\psi + \omega\Delta t) + \cos(\psi)) \quad (9)$$

$$v(k+1) = v(k) \quad (10)$$

$$\psi(k+1) = \psi(k) + \omega\Delta t \quad (11)$$

$$\omega(k+1) = \omega(k) \quad (12)$$

When a target undergoes linear motion where the angular velocity is close to zero ($\omega \rightarrow 0$), a straight-line linear approximation is used to avoid division-by-zero numerical instabilities[cite: 1]:

$$p_x(k+1) = p_x(k) + v \cos(\psi)\Delta t \quad (13)$$

$$p_y(k+1) = p_y(k) + v \sin(\psi)\Delta t \quad (14)$$

4.2 Measurement Mapping Function

Because the processed LiDAR clusters directly yield spatial position coordinates in the Cartesian plane, the observation function $h(x)$ mapping state parameters to measurement space remains completely linear[cite: 1]:

$$z_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} x_k + v_k \quad (15)$$

5 Algoritmik Detaylarıyla Unscented Kalman Filter (UKF)

The implementation steps of the UKF cycle are executed sequentially according to the following mathematical formulations[cite: 1]:

5.1 Sigma Point Generation

Given an n -dimensional state vector ($n = 5$) with an estimated mean $\hat{x}_{k|k}$ and an error covariance matrix $P_{k|k}$, a set of $2n + 1$ deterministic sigma points χ_i is constructed around the mean:

$$\chi_{0,k|k} = \hat{x}_{k|k} \quad (16)$$

$$\chi_{i,k|k} = \hat{x}_{k|k} + \left(\sqrt{(n + \lambda)P_{k|k}} \right)_i, \quad i = 1, \dots, n \quad (17)$$

$$\chi_{i+n,k|k} = \hat{x}_{k|k} - \left(\sqrt{(n + \lambda)P_{k|k}} \right)_i, \quad i = 1, \dots, n \quad (18)$$

where $\lambda = \alpha^2(n + \kappa) - n$ is a scaling parameter determining the spread of points, α controls primary scaling, and κ is a secondary tuning parameter. Corresponding mean and covariance weights (W^m, W^c) are defined by:

$$W_0^m = \frac{\lambda}{n + \lambda}, \quad W_0^c = \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta) \quad (19)$$

$$W_i^m = W_i^c = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n \quad (20)$$

5.2 Time Update (Prediction)

Every generated sigma point is propagated directly through the non-linear CTRV kinematic function $f(\cdot)$ [cite: 1]:

$$\chi_{i,k|k-1} = f(\chi_{i,k|k}) \quad (21)$$

The predicted state mean $\hat{x}_{k|k-1}$ and predicted error covariance $P_{k|k-1}$ are formed via weighted summation[cite: 1]:

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2n} W_i^m \chi_{i,k|k-1} \quad (22)$$

$$P_{k|k-1} = \sum_{i=0}^{2n} W_i^c (\chi_{i,k|k-1} - \hat{x}_{k|k-1}) (\chi_{i,k|k-1} - \hat{x}_{k|k-1})^T + Q \quad (23)$$

5.3 Measurement Update (Correction)

Predicted states are transformed into measurement sigma points $\mathcal{Z}_{i,k|k-1}$ via the observation model[cite: 1]:

$$\mathcal{Z}_{i,k|k-1} = h(\chi_{i,k|k-1}) \quad (24)$$

The expected observation mean $\hat{z}_{k|k-1}$ and innovation covariance matrix S_k are evaluated as follows[cite: 1]:

$$\hat{z}_{k|k-1} = \sum_{i=0}^{2n} W_i^m \mathcal{Z}_{i,k|k-1} \quad (25)$$

$$S_k = \sum_{i=0}^{2n} W_i^c (\mathcal{Z}_{i,k|k-1} - \hat{z}_{k|k-1}) (\mathcal{Z}_{i,k|k-1} - \hat{z}_{k|k-1})^T + R \quad (26)$$

Computing the cross-covariance matrix $P_{xz,k}$ allows calculation of the optimal Kalman gain K_k [cite: 1]:

$$P_{xz,k} = \sum_{i=0}^{2n} W_i^c (\chi_{i,k|k-1} - \hat{x}_{k|k-1}) (\mathcal{Z}_{i,k|k-1} - \hat{z}_{k|k-1})^T \quad (27)$$

$$K_k = P_{xz,k} S_k^{-1} \quad (28)$$

Finally, the updated state estimate $\hat{x}_{k|k}$ and posterior error covariance $P_{k|k}$ are obtained using the innovation residual vector $\nu_k = z_k - \hat{z}_{k|k-1}$ [cite: 1]:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k \quad (29)$$

$$P_{k|k} = P_{k|k-1} - K_k S_k K_k^T \quad (30)$$

6 Data Association and Track Management

To maintain tracking continuity for multiple targets simultaneously, a nearest-neighbor data association framework is implemented[cite: 1]. For every active track, a validation gate boundary based on the Mahalanobis distance is defined[cite: 1]:

$$d_{gate} = \sqrt{(z_k - \hat{z}_{k|k-1})^T S_k^{-1} (z_k - \hat{z}_{k|k-1})} \quad (31)$$

An observation vector is considered valid for assignment if it satisfies the constraint[cite: 1]:

$$d_{gate} < \gamma \quad (32)$$

where γ is a tunable chi-squared validation gate threshold[cite: 1].

To handle environmental clutter and target occlusions, explicit track management logic is deployed[cite: 1]:

- **Track Initialization:** Unassociated observations trigger a temporary candidate track. If the target is successfully detected across a consecutive number of frames (confirmation logic), the track status updates to *Active*[cite: 1].
- **Track Deletion (Missed-Frame Handling):** When a target is temporarily blocked or moves out of the line of sight, the UKF continues to predict the target's trajectory without measurement updates[cite: 1]. If the track fails to associate with a measurement for a maximum number of consecutive frames, it is dropped from the active track list to minimize false tracks[cite: 1].

7 Filter Consistency Analysis

As highlighted in **EE4084_Project_Paper.pdf**, tracking performance and stability are assessed via Normalized Innovation Squared (NIS) metrics[cite: 1]:

$$NIS = \nu_k^T S_k^{-1} \nu_k \quad (33)$$

Given a measurement space dimension of $m = 2$, the calculated NIS values follow a Chi-square distribution $\chi^2(m)$ [cite: 1]. A 95% statistical confidence interval bound is enforced to evaluate if the filter is operating correctly[cite: 1]:

$$\chi_{low}^2 < NIS < \chi_{high}^2 \quad (34)$$

If the NIS continuously scales above the high bound, it indicates under-estimated process noise covariances or unmodeled maneuvering forces[cite: 1]. Conversely, values lingering below the lower bound suggest over-conservative measurement uncertainties[cite: 1].

8 Project Management, Timeline, and Deliverables

The project will be completed across distinct development phases, ensuring comprehensive verification of hardware interfaces and estimation software.

Table 1: Proposed Project Development Timeline

Phase	Core Tasks & Objectives	Duration
Phase 1	Hardware interface configuration and raw polar telemetry extraction	Weeks 1–2
Phase 2	DBSCAN clustering implementation and centroid identification	Weeks 3–4
Phase 3	UKF framework programming with CTRV motion dynamics	Weeks 5–7
Phase 4	Gating and track management implementation	Weeks 8–9
Phase 5	Quantitative verification (RMSE, NIS) and final adjustments	Weeks 10–12

8.1 Expected Final Deliverables

- **Software Stack:** A unified Python and MATLAB execution workspace containing the DBSCAN extraction script, recursive UKF loop, and data association modules[cite: 1].
- **Comprehensive Evaluation:** Empirical graphs showcasing multi-target trajectories, tracking continuity over occlusions, and real-time processing latencies per frame[cite: 1].

9 Conclusion

This proposal details an integrated hardware-software framework for real-time multi-target tracking using a YDLIDAR G2 sensor and an Unscented Kalman Filter paired with a CTRV motion model[cite: 1]. By addressing the non-linear limitations of traditional estimation techniques and utilizing a robust DBSCAN clustering backend, the proposed

framework provides stable trajectory estimation in cluttered indoor environments[cite: 1]. Validation through NIS consistency checks ensures predictable filter performance, making this architecture a flexible solution for modern autonomous robotics and intelligent indoor navigation[cite: 1].

References

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